7.05 Decision Making in Project Management
Networks are applied to decision making, in particular to activity networks and critical path analysis, including scheduling.

should be used to demonstrate the power of these algorithms in large scale problems; in the assessment, learners will be asked to demonstrate their understanding of the application of algorithms to small scale problems only.

Use of technology

To support the teaching and learning of mathematics using technology, we suggest that the following activities are carried out through the course:

1. Graphing tools: Learners could use graphing software to perform graphical linear programming, and to investigate the effects on the solution of changing coefficients and parameters.

2. Networks and network algorithms: Learners could use software to investigate networks and to implement network algorithms, in particular for networks which are too large to work with by hand.

3. Algorithms: Learners could use spreadsheets or a suitable programming language to implement simple algorithms and to understand how to create algorithms to perform simple tasks.

4. Computer Algebra System (CAS): Learners could use CAS software to draw and manipulate graphs, to explore algebraic relationships. This is best done in conjunction with other software such as graphing tools and spreadsheets.

5. Simulation: Learners could use spreadsheets to simulate contexts in game theory, including investigating the long term effects of particular strategies.

Content of Discrete Mathematics (Optional paper Y544)

When this course is being co-taught with AS Level Further Mathematics A (H235) the ‘Stage 1’ column indicates the common content between the two specifications and the ‘Stage 2’ column indicates content which is particular to this specification. In each section the OCR reference lists ‘Stage 1’ statements before ‘Stage 2‘ statements.

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
7.01 Mathematical Preliminaries			
7.01a	Types of problem	a) Understand and be able to use the terms “existence’” “construction”, “enumeration” and “optimisation” in the context of problem solving. *Includes classifying a given problem into one or more of these categories.*	
7.01b	Set notation	b) Understand and be able to use the basic language and notation of sets. *Includes the term “partition” and counting the number of partitions of a set including with constraints.*	
7.01c	The pigeonhole principle	c) Be able to use the pigeonhole principle in solving problems.	

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
7.01d	Arrangement and selection problems	d) Understand and use the multiplicative principle. Includes knowing that the number of arrangements of n distinct objects is $\prod_{r=1}^{n} r = n!$.	
7.01e		e) Be able to enumerate the number of ways of obtaining an ordered subset (permutation) of r elements from a set of n distinct elements. *Includes using the notation* ${}_n\mathrm{P}_r = {}^n\mathrm{P}_r$.	
7.01f		f) Be able to enumerate the number of ways of obtaining an unordered subset (combination) of r elements from a set of n distinct elements. *Includes using the notation* ${}_n\mathrm{C}_r = {}^n\mathrm{C}_r$.	
7.01g 7.01h		g) Be able to solve problems about enumerating the number of arrangements of objects in a line, including those involving: 1. repetition, *e.g. how many different eight digit numbers can be made from the digits of 12333210?* 2. restriction, *e.g. how many different eight digit numbers can be made from the digits of 12333210 if the two 2s cannot be next to each other?*	h) Be able to solve problems about enumerating the number of arrangements of only some of a group of objects. *e.g. how many different four digit numbers can be made from the digits of 12333210?* *e.g. how many different numbers greater than 20 000 can be made from the digits of 12333210?*
7.01i 7.01j		i) Be able to solve problems about selections, including with constraints. *e.g. Find the number of ways in which a team of 3 men and 2 women can be selected from a group of 6 men and 5 women.*	j) Be able to solve problems with several constraints. *e.g. Given a graph showing who dislikes who, find the number of ways of choosing 3 men and 2 women from a group of 4 men and 4 women so that no two people chosen dislike each other.*

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
7.01k 7.01l 7.01m	The inclusion-exclusion principle	k) Be able to use the inclusion-exclusion principle for two sets in solving problems. *e.g.* $n(A \cup B) = n(A) + n(B) - n(A \cap B)$. *Venn diagrams may be used.* *e.g. How many integers in* $\{1, 2, ..., 100\}$ *are not divisible by* 2 *or* 3*?*	l) Be able to extend the inclusion-exclusion principle to more than two sets. *e.g.* $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$. *Venn diagrams may be used.* *e.g. How many integers in* $\{1, 2, ..., 100\}$ *are not divisible by* 2, 3 *or* 5*?* m) Be able to find derangements. *Includes enumeration of the number of derangements of* n *objects,* D_n*, by ad hoc methods only.*
7.02 Graphs and Networks			
7.02a 7.02b 7.02c	Terminology and notation	a) Understand the meaning of the terms "vertex" (or "node") and "arc" (or "edge"). *Includes the concept of the "degree" of a vertex as the number of arcs "incident" to the vertex.* *Includes the term "adjacent" for pairs of vertices or edges.* b) Understand the meaning of the terms "tree", "simple", "connected" and "simply connected" as they refer to graphs. *Includes understanding and using the restrictions on the vertex degrees implied by these conditions.* c) Understand the meaning of the terms "walk", "trail", "path", "cycle" and "route". *A "walk" is a set of arcs where the end vertex of one is the start vertex of the next.* *A "trail" is a walk in which no arcs are repeated.* *A "path" is a trail in which no nodes are repeated.* *A "cycle" is a closed path.* *A "route" can be a walk, a trail or a path, or may be a closed*	

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
7.02d	Complete graphs	d) Understand and be able to use the term "complete" and the notation K_n for a complete graph on n vertices. *Includes knowing that K_n has $\frac{1}{2}n(n-1)$ arcs.*	
7.02e 7.02f	Bipartite graphs	e) Understand and be able to use bipartite graphs and the notation $K_{m,n}$ for a complete bipartite graph connecting m vertices to n vertices. *Includes knowing that $K_{m,n}$ has mn arcs.*	f) Use a colouring argument to show that a given graph is, or is not, bipartite.
7.02g	Eulerian graphs	g) Use the degrees of vertices to determine whether a given graph is Eulerian, semi-Eulerian or neither. Understand what these terms mean in terms of traversing the graph.	
7.02h 7.02i	Hamiltonian graphs		h) Understand and be able to use the definition of a Hamiltonian path, a Hamiltonian cycle and a Hamiltonian graph. i) Know and use Ore's theorem. *i.e. For a simple graph G with $n \geq 3$ vertices, if $\deg v + \deg w \geq n$ for every pair of distinct non-adjacent vertices v and w, then G is Hamiltonian.* *Includes understanding that Ore's theorem gives a sufficient but not necessary condition for a graph to be Hamiltonian.*

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
7.02j	Isomorphism	j) Understand what it means to say that two graphs are isomorphic. Construct an isomorphism either by a reasoned argument or by explicit labelling of vertices. *Includes understanding that having the same degree sequence (ordered list of vertex degrees) is necessary but not sufficient to show isomorphism.* *Includes the term "non-isomorphic".*	
7.02k	Digraphs	k) Understand and be able to use digraphs. *Includes the terms "indegree" and "outdegree".*	
7.02l	Planar graphs		l) Understand and be able to apply the concepts of planarity, subdivision and contraction. *i.e. Subdivision is inserting a vertex of degree 2 into an arc. Contraction is contracting two vertices into one so that any arc incident with the original two vertices is incident with the contracted vertex.* *Includes the notation (AB) for the vertex created by the contraction of the arc AB.* *Includes drawing planar representations.*
7.02m			m) Know, understand and use Euler's formula $\mathrm{V} + \mathrm{R} = \mathrm{E} + 2$.
7.02n			n) Know and use Kuratowski's theorem. *i.e. A finite graph is planar if and only if it does not contain a subgraph that is a subdivision of K_5 or $K_{3,3}$.*
7.02o			o) Understand and be able to use the concept of thickness. [*Calculation of thickness > 2 is excluded.*]

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
7.02p 7.02q 7.02r	Using graphs and networks	p) Understand that a network is a weighted graph. Use graphs and networks to model the connections between objects. *Graphs and networks may be directed or undirected.* q) Use an adjacency matrix representation of a graph and a weighted matrix representation of a network. r) Be able to model problems using graphs or networks, and solve them.	
7.03 Algorithms			
7.03a	Definition of an algorithm	a) Understand that an algorithm has an input and an output, is deterministic and finite. *Includes the use of a counter and the use of a stopping condition in an algorithm.* *Be familiar with the terms "greedy", "heuristic" and "recursive" in the context of algorithms.*	
7.03b	Awareness of the uses and practical limitations of algorithms	b) Appreciate why an algorithmic approach to problem-solving is generally preferable to ad hoc methods, and understand the limitations of algorithmic methods. *Includes understanding that algorithmic methods are used by computers for solving large scale problems and that small scale problems are only being used to demonstrate how a given algorithm works.*	

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
7.03c	Working with algorithms	c) Trace through an algorithm and interpret what the algorithm has achieved. Algorithms may be presented as flow diagrams, listed in words, or written in simple pseudo-code. *Includes understanding and being able to use the functions* $\mathrm{INT}(x)$ *and* $\mathrm{ABS}(x)$*. Learners may find it useful to have a calculator with these functions, but large numbers of repeated applications will not be required in the assessment.* *Includes adapting or altering an algorithm to achieve a given purpose, and adjusting a short set of instructions to create an algorithm.* [*Programming skills will not be required*.]	
7.03d	The order of an algorithm	d) Use the order of an algorithm to calculate an approximate run-time for a large problem by scaling up a given run-time. *Includes understanding that when the "maximum run-time" of an algorithm is represented as a function of the "size" of the problem, the order of the algorithm, for very large sized problems, is given by the dominant term.* *Learners should know that the sum of the first* n *positive integers is* $\sum_{r=1}^{n} r = \frac{1}{2}n(n+1)$. *Learners should be familiar with the notation* $\mathrm{O}(n^4)$ *and the concept of dominance in an informal sense only.*	

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
7.03e	Efficiency and complexity	e) Compare the efficiency of two algorithms that achieve the same end result by considering a given aspect of the run-time in a specific case. *e.g. The number of swaps or comparisons to sort a given list*.	
7.03f 7.03h		f) Calculate worst case time complexity, the “maximum run-time” $\mathrm{T}(n)$, as a function of the size of a problem by considering the worst case for a specific problem. *Includes cases of the algorithms for sorting and standard network problems studied in this specification*. *Includes an informal understanding that, for example* $\mathrm{T}(n) = n^2 + n^4$ *is order* n^4*, or equivalently* $\mathrm{O}(n^4)$.	h) Calculate the run-time as a function of the size by considering the best case, the worst case or *Includes considering all cases and averaging where a*
7.03g 7.03i		g) Be familiar with $\mathrm{O}(n^k)$, where *n* is a measure of the size of the problems and $k = 0,\ 1,\ 2,\ 3$ or 4.	i) Be familiar with: $\mathrm{O}(n^k)$ for $k \in \mathbb{R}$, $\mathrm{O}(a^n)$ for $a > 0$, $\mathrm{O}(\log n)$ where n is a measure of the size of th *Know the hierarchy of orders and what this means in efficiency. Learners should be aware that:* $\mathrm{O}(1) \subset \mathrm{O}(\log n) \subset \mathrm{O}(n) \subset \mathrm{O}(n\log n) \subset \mathrm{O}(n^2) \subset \mathrm{O}(n$ $\subset \mathrm{O}(a^n) \subset \mathrm{O}(n!)$, *which is given in the Formulae Boo*

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
7.03j 7.03k	Strategies for sorting	j) Be able to sort a list using bubble sort and using shuttle sort. *Bubble sort and shuttle sort will start at the left-hand end of the list, unless specified otherwise in the question.* *Includes knowing that, in general, sorting algorithms have quadratic order as a function of the length of the list.*	k) Be able to sort a list using quick sort. *Quick sort will pivot on the first value, unless specified otherwise in a question.* *Includes knowing that quick sort is only* $\mathrm{O}(n^2)$ *in the worst case.* *Questions may be set that interrogate the application of the method, for example whether the choice of pivot affects the efficiency of quick sort.*
7.03l 7.03m	Strategies for packing	l) Be familiar with the next-fit, first-fit, first-fit decreasing and full bin methods for one-dimensional packing problems. *Includes knowing that these are heuristic algorithms.* *Includes the terms "online" and "offline".*	m) Extend their knowledge of packing methods. *e.g. Packing problems in two or three dimensions.* *e.g. Knapsack problems: given a set of items, each with a mass and a profit, determine which to include so that the total mass does not exceed some given limit and the total profit is as large as possible.*

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
7.04 Network Algorithms			
7.04a	Least weight path between two vertices	a) Be able to use examples to demonstrate understanding and use of Dijkstra's algorithm to find the length and route of a least weight (shortest) path. *Solve problems that require a least weight (shortest) path as part of their solution.* *Know that Dijkstra's algorithm has quadratic order (as a function of the number of vertices).*	
7.04b	Least weight set of arcs connecting all vertices	b) Be able to use examples to demonstrate understanding and use of Prim's algorithm (both in graphical and tabular/matrix form) and Kruskal's algorithm to find a minimum connector (minimum spanning tree) for a network. *Solve problems that require a minimum spanning tree as part of their solution.* *Includes adapting a solution to deal with practical issues.* *Know that Prim's algorithm and Kruskal's algorithm have cubic order (as a function of the number of vertices).*	

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
7.04c	Least weight cycle through all vertices		c) Be able to use the nearest neighbour method on a network formed by weighting a complete graph to find an upper bound for the travelling salesperson problem. *Includes understanding that when the nearest neighbour method is used on a network formed by weighting a graph that is not complete it may stall before reaching every vertex or it may reach every vertex but to close the route it may need a path that is not a direct connection from the end vertex back to the start vertex.* *Includes choosing between two, or more, upper bounds to find the best (least) upper bound.* *Includes using short-cuts where possible to improve an upper bound.*
7.04d			d) Be able to use a minimum spanning tree on a reduced network to calculate a lower bound for the travelling salesperson problem on a complete graph and understand why this method gives a lower bound. *Understand that for a graph that is not complete this method can give a value that is not a lower bound.* *Includes choosing between two or more lower bounds to find the best (greatest) lower bound.*

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
7.04e	Least weight route through all vertices that traverses every arc at least once		e) Be able to solve the route inspection problem by consideration of all possible pairings of up to six odd nodes. *If a problem has more than six odd nodes additional restrictions will reduce the number of pairings that need to be considered.* *Problems may be set that require an understanding of how the number of pairings increases as the number of odd nodes increases. For* $2n$ *odd nodes the number of pairings is* $1 \times 3 \times \ldots \times (2n-1) = \prod_{r=1}^{n}(2r-1)$.
7.04f	Network problems	f) Be able to choose an appropriate algorithm to solve a practical problem. *Includes adapting an algorithm or a solution to deal with practical issues.*	

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…
7.05 Decision Making in Project Management			
7.05a	Critical path analysis	a) Be able to construct and interpret activity networks using activity on arc. *Appreciate that a path of critical activities (a critical path) is a longest path in a directed network.*	
7.05b		b) Be able to carry out a forward pass to determine earliest start times and find the minimum project completion time, and to carry out a backward pass to determine latest finish times and find the critical activities. *Includes understanding and using the terms "burst" and "merge".*	
7.05c		c) Understand, and be able to calculate, (total) float.	
7.05d			d) Be able to find latest start times and earliest finish times for activities. Calculate and interpret independent and interfering float.
7.05e			e) Be able to use an activity network to construct a cascade chart and use a cascade chart to determine the effect on the minimum project completion time of a delay to one or more activities, or other scheduling restrictions. *Cascade charts may be constructed either with one activity on each row or with the critical activities together in one row and a row for each non-critical activity.* *Float may be shown using dashed lines.* *Includes constructing a schedule to show how a given number of workers can complete a project subject to given constraints.*

OCR Ref.	Subject Content	Stage 1 learners should…	Stage 2 learners should additionally…